

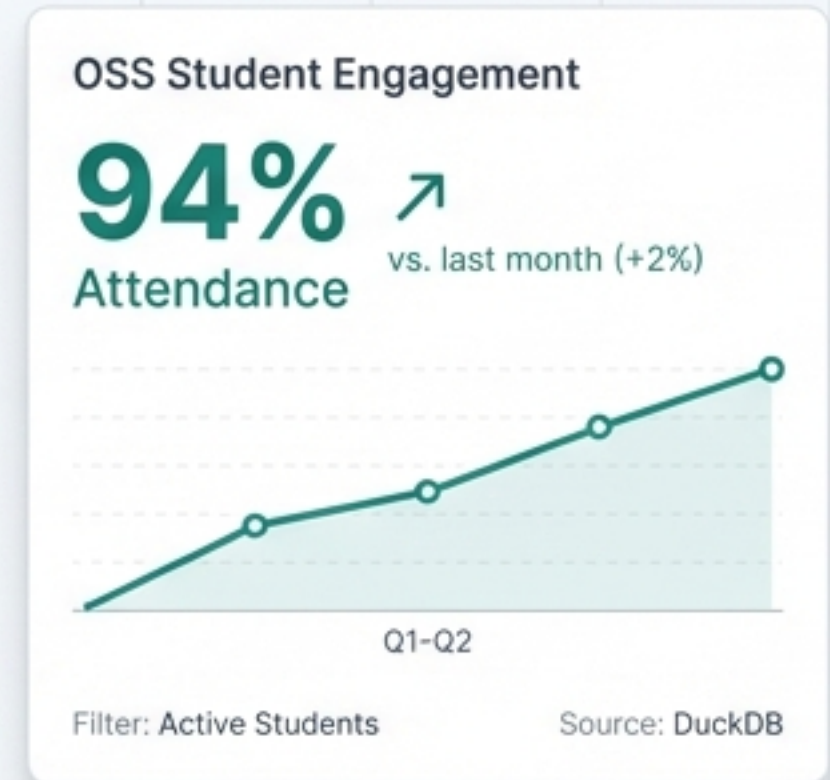
Metabase for OSS Student Analytics

From Raw Data to Intervention

A **technical guide** to querying, modeling, and visualizing student data using the DuckDB-Metabase stack.



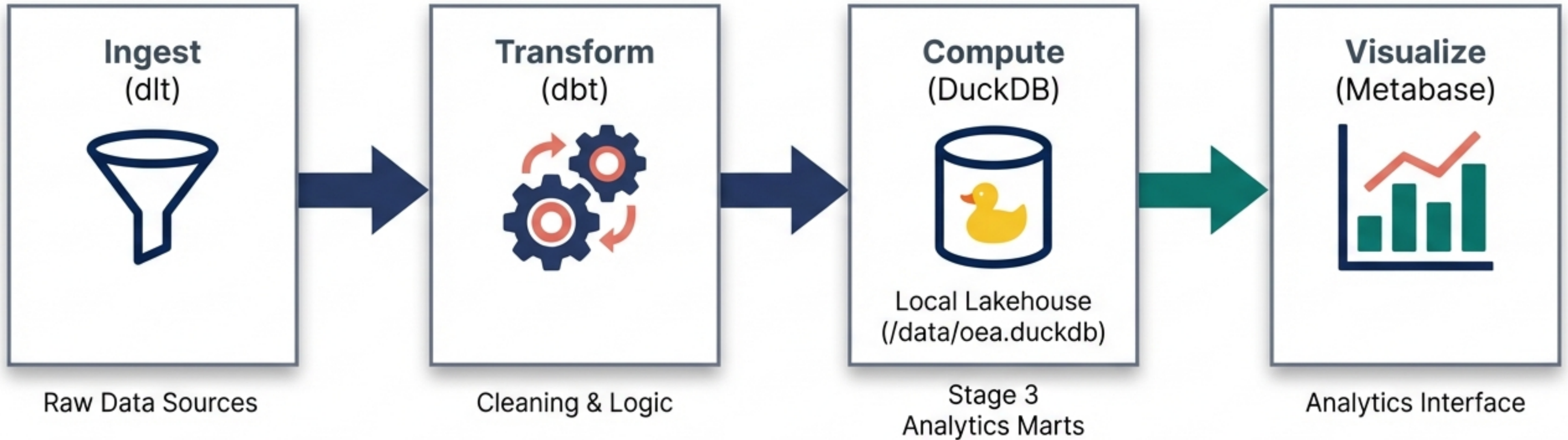
CSV/Raw Data



Card

Context: Stage 3 Analytics Consumption Layer | OSS Framework

The Architecture: Where Metabase Fits in the Lakehouse



The Data Lakehouse

Metabase connects directly to the local DuckDB database file. Unlike traditional BI requiring complex warehousing, this allows for **sub-second** queries on local files while maintaining a Medallion architecture (Bronze/Silver/Gold).

Data Freshness

Views are refreshed nightly via dbt schedules (typically 02:00 AM), ensuring dashboards reflect the previous day's attendance and grade data.

Interface Orientation & The 'Stage 3' Environment

The screenshot shows a 'Browse Data' interface with a left sidebar and a main content area. The sidebar contains three items: 'Our Analytics' (selected), 'Personal Collection', and 'Your Saved Questions'. The main content area displays the 'Schema: main_main_analytics' with a list of four views, each with a grid icon and a number to its left. The first view, 'v_chronic_absenteeism_risk', is highlighted with a yellow background. Three callouts with numbered circles point to specific elements: Callout 1 points to the 'Our Analytics' folder; Callout 2 points to the 'v_chronic_absenteeism_risk' view; Callout 3 points to the 'v_chronic_absenteeism_risk' view.

Browse Data

- Our Analytics
- Personal Collection
- Your Saved Questions

Schema: main_main_analytics

- 1 v_chronic_absenteeism_risk
- 2 v_wellbeing_risk_profiles
- 3 v_equity_outcomes
- 4 v_class_section_comparison

1 Master Library: The official folder structure for the district.

2 Production Views: These are Stage 3 assets. Clean, joined, and ready for reporting.

3 Key Asset: Contains student-level risk scores and 30-day attendance rates.

The Query Builder: Answering Questions Without SQL

The GUI replaces the need for writing SELECT ... GROUP BY.



Result Preview

Homeroom Teacher	Count
Smith, J.	24
Doe, A.	18
Rivera, M.	42

Workflow: This sequence instantly identifies which teachers have the highest density of at-risk 9th graders.

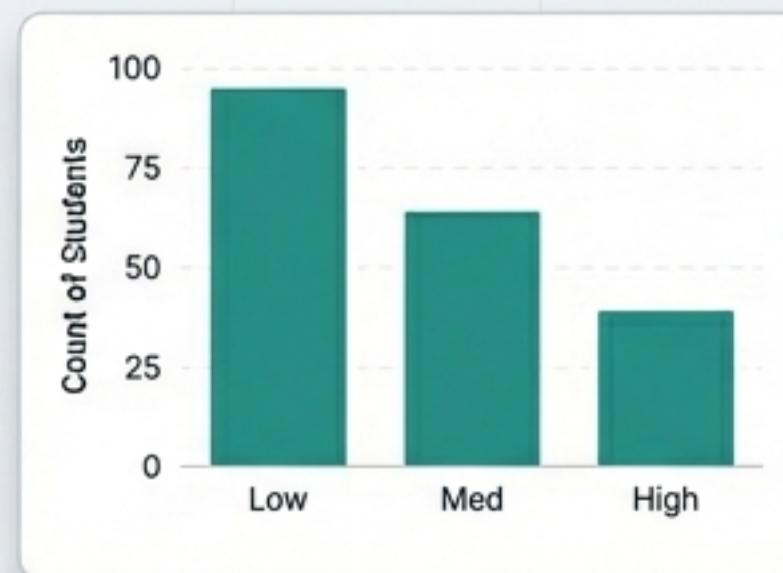
Visualizing Student Data: Choosing the Right Chart



Trend / Line Chart

Use for time-series.

Example: Attendance Rate over last 6 months.

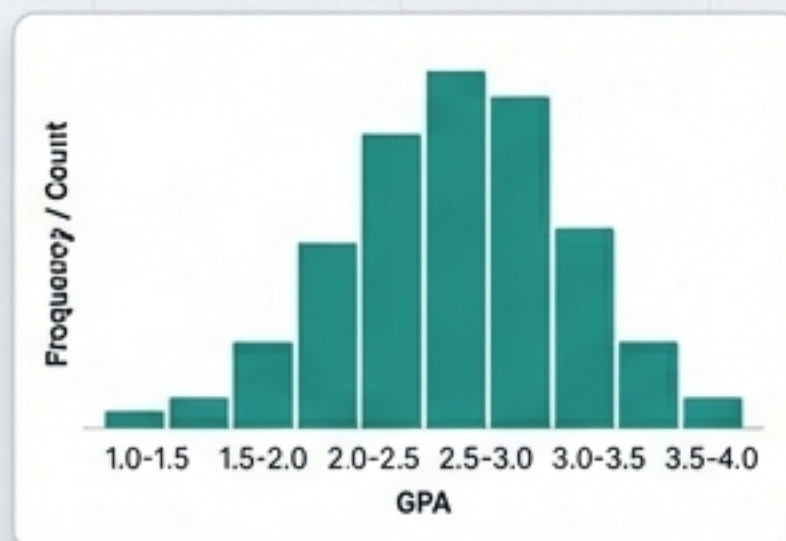


Bar Chart

Use for categorical comparisons.

Example: Count of students by Risk Level.

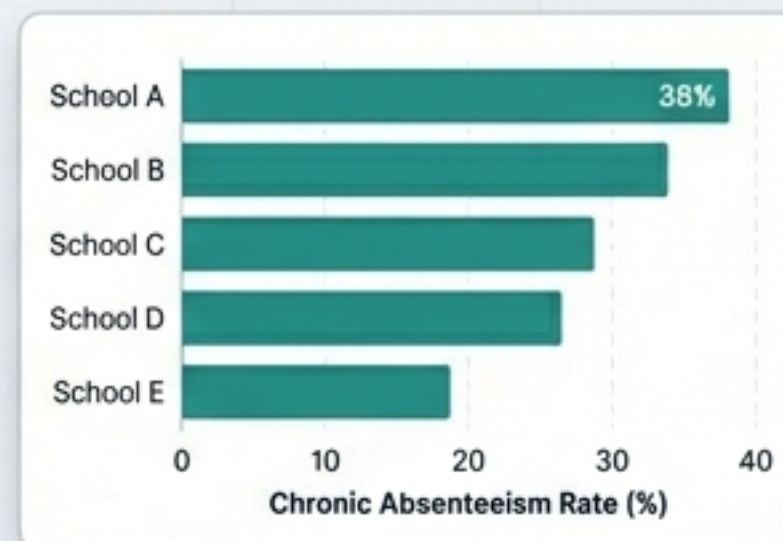
Histogram



Use for distributions.

Example: GPA distribution across the district.
Metabase automatically "bins" continuous data.

Row Chart



Use for ranking many categories.

Example: Ranking 30 schools by Chronic Absenteeism Rate.

Joining Data: From VLOOKUP to Relational Models

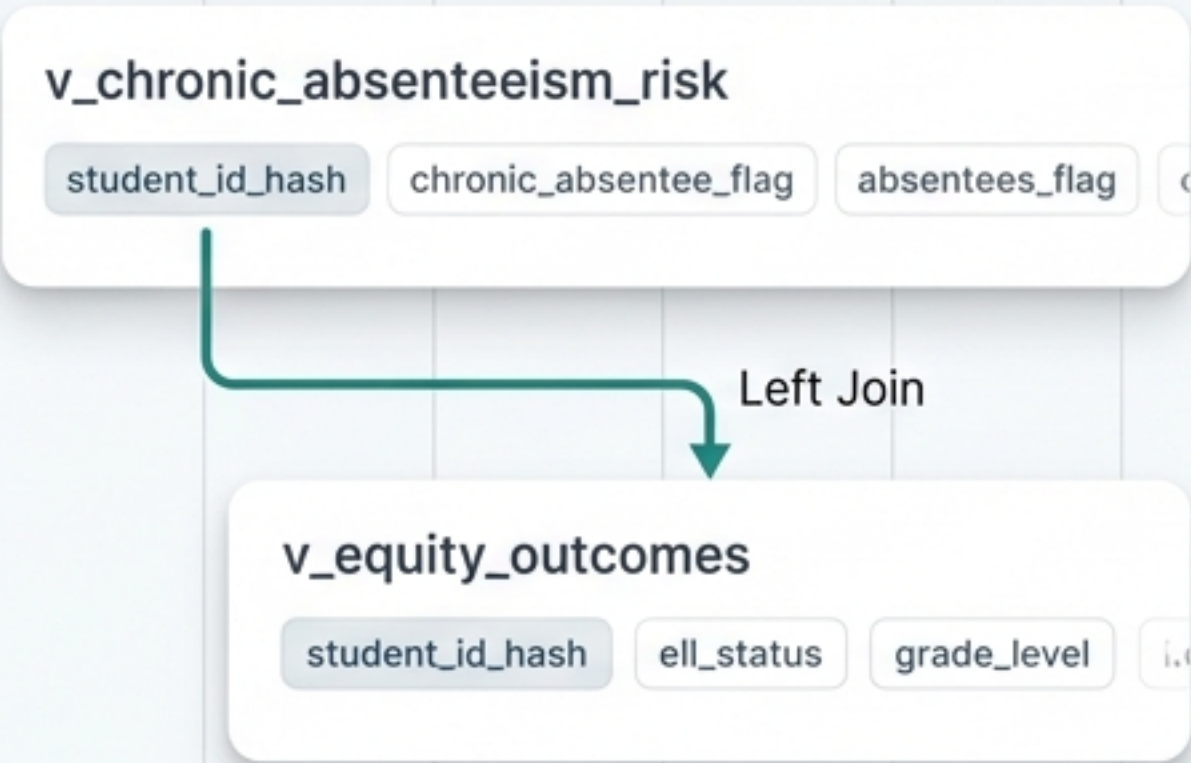
The Excel Mental Model

	A	B	C
1			
2	3001840531	Present	=XLOOKUP(A2, AttendanceA:A, AttendanceB:B)
3			
4			
5			
6			
7			
8			

Pulling data
across sheets



The Metabase Mental Model



Join on shared key to analyze
Absenteeism by ELL Status.

Pro Tip: Prefer 'Left Joins' to keep all students from your primary list, even if they lack data in the secondary table.

Custom Expressions: Feature Engineering in the GUI

Create a custom column

Field name

Risk Status

Formula

`case([Attendance Rate] < 0.9, 'Chronically Absent', 'On Track')`

Done

Categorization (Case)

Convert raw numbers into status labels. Useful for creating binary filters like 'On Track' vs 'Off Track'.

```
case([Attendance Rate] < 0.9,  
     'Chronically Absent', 'On Track')
```

Math Operations

Calculate gaps or ratios. Visualizes performance relative to the mean.

```
[GPA] - [District Average GPA]
```

Text Cleaning

Format names or codes for reports. Formula: `concat([Last Name], ', ', [First Name])`

```
concat([Last Name], ', ', [First Name])
```


The SQL Editor: Native Querying & Variables

Using variables to make static SQL interactive.

```
SELECT *  
FROM v_wellbeing_risk_profiles  
WHERE grade_level = {{Grade}}
```

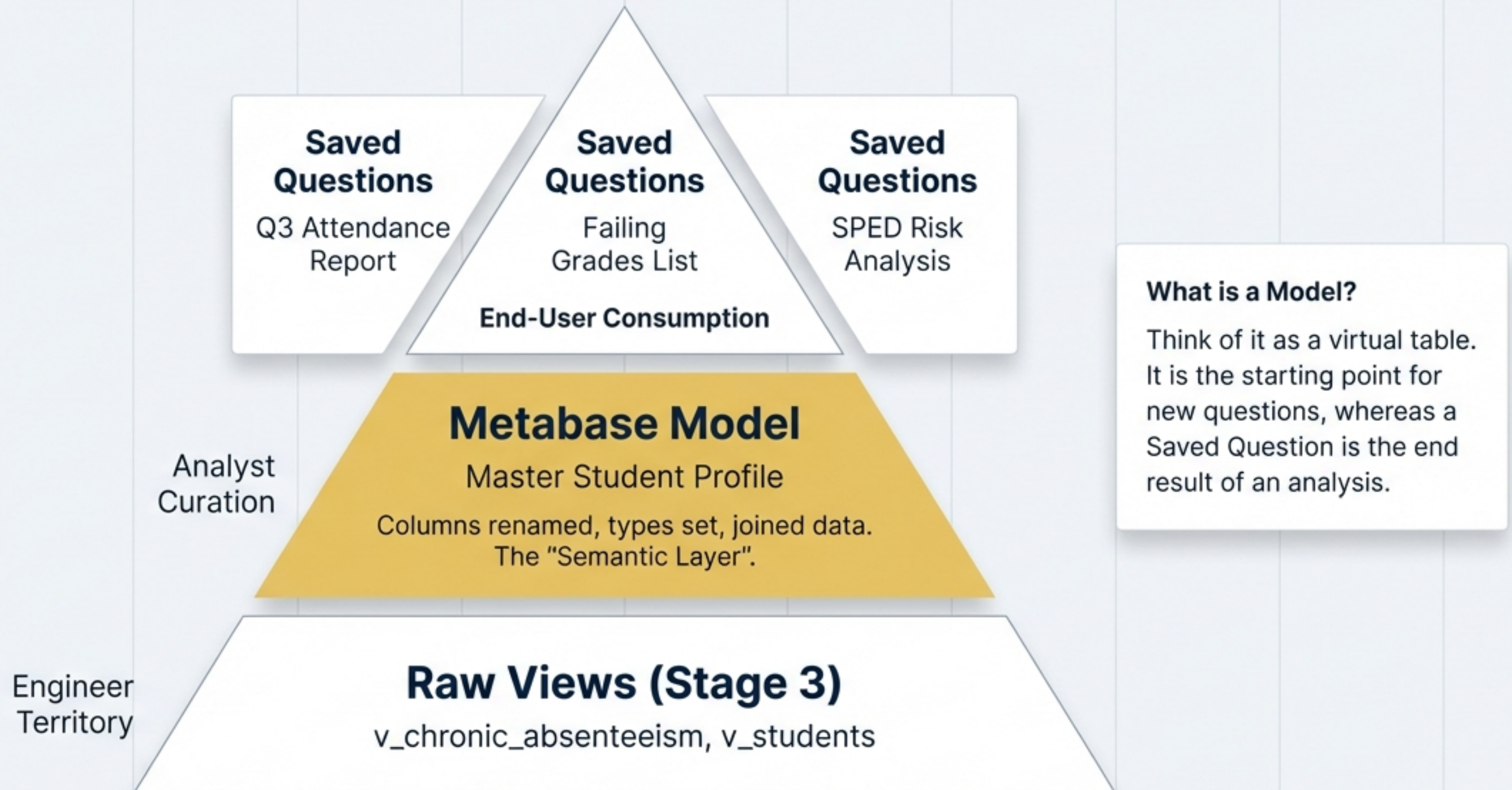
Variable Sidebar: Variable
Name: Grade | Type: Text

Grade

This code generates this widget. Users pick '10' from the dropdown, and the SQL executes dynamically.

Field Filters: A special variable type that maps to the Data Model, allowing smart dropdowns (e.g., a list of all School Names) instead of manual text entry.

Models vs. Saved Questions: Curating the Experience



Metadata & Curation: Gardening the Data

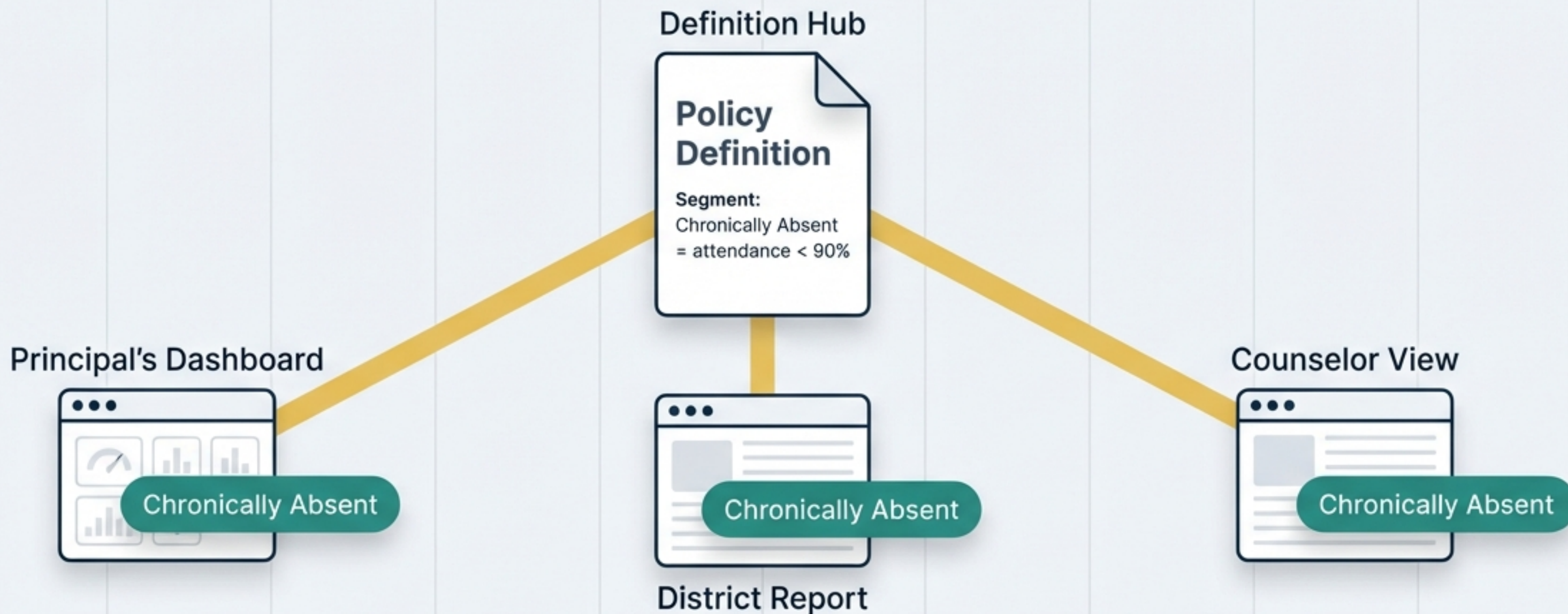
Field Name	Type	Settings/Visibility
risk_score	Number	Formatting: 0 decimal places
profile_url	URL	
_dlt_load_id	Text	Do not include in results ☐ OFF

Impact: Makes the text clickable in tables to open external profiles.

Hide technical debt to reduce cognitive load.

Explicitly setting types and hiding technical columns transforms a raw database into a user-friendly product.

Segments & Metrics: Defining a Shared Language



Why? If district policy changes to 88%, you update the Segment definition once centrally, and all downstream dashboards update automatically. It establishes a **Single Source of Truth**.

Dashboard Design: Hierarchy and Clarity

KPIs: Above the Fold

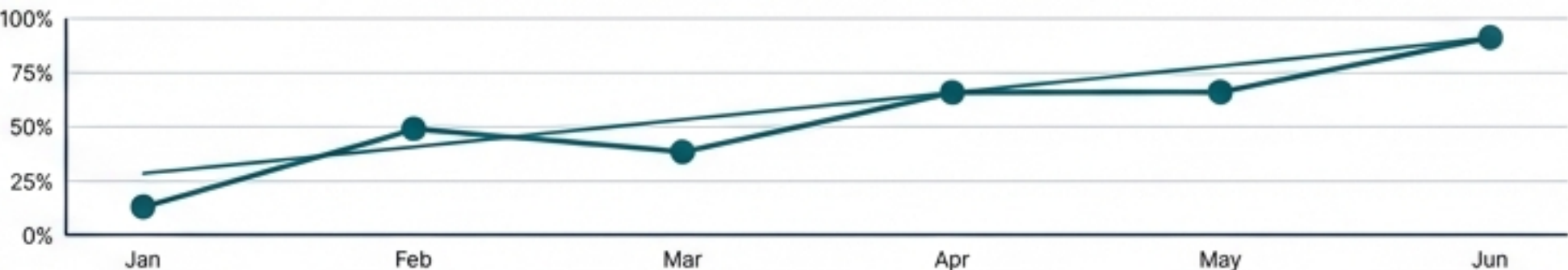
Active Students:
12,405

District Attendance:
94.2%

Students at Risk:
450

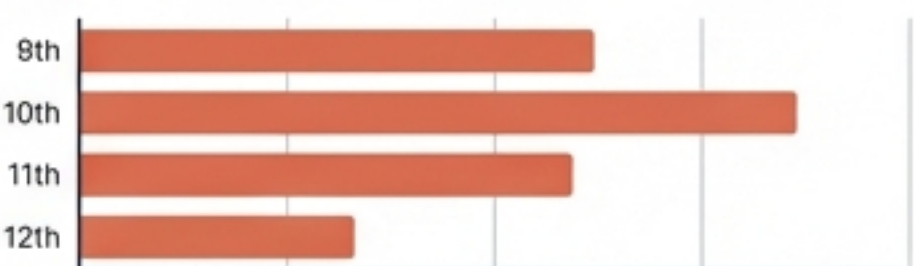
Context & Trends

Attendance Trend (6 Months)



Breakouts

Risk by Grade Level



Schools List

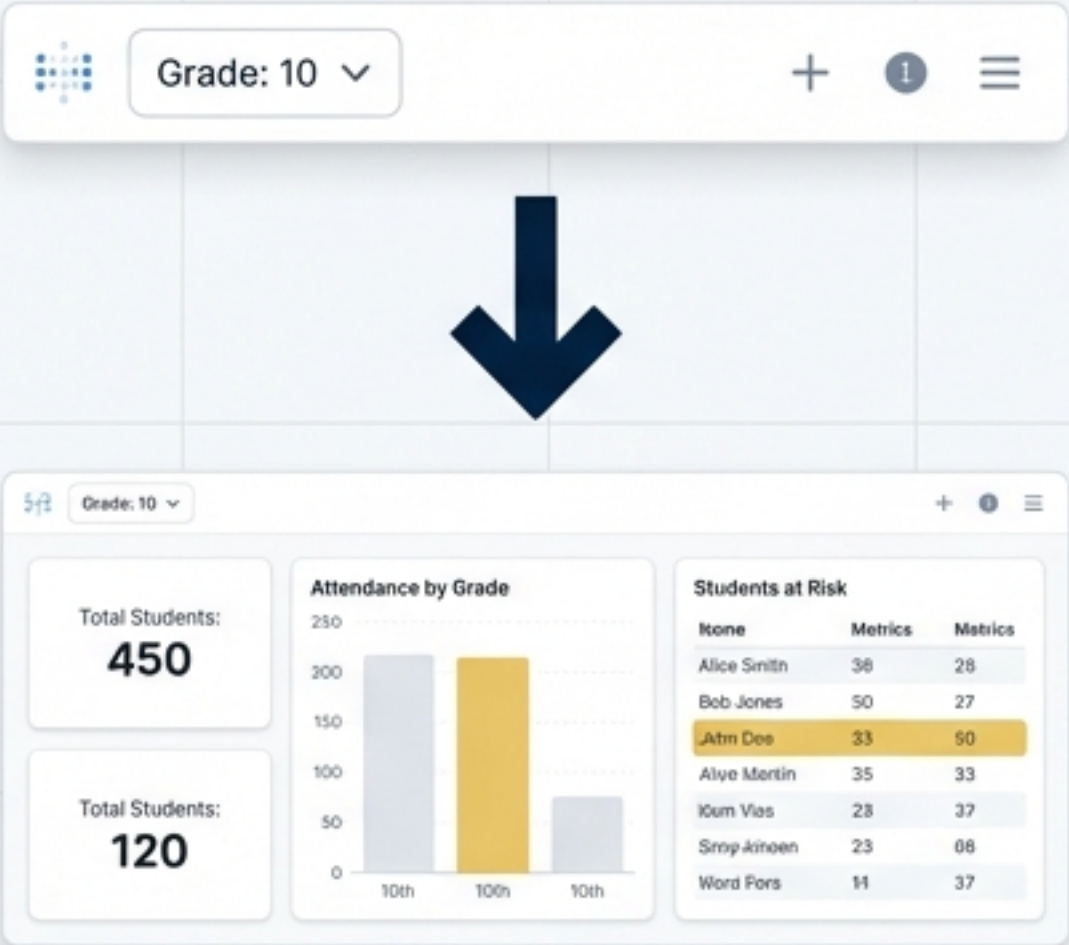
School Name	Risk Count	Attendance Rate
Lincoln High	120	93%
Lincoln High	29	96%
Lincoln High	10	96%

Grouping: Use Text Cards as headers (e.g., '## Academic Performance') to create visual sections.

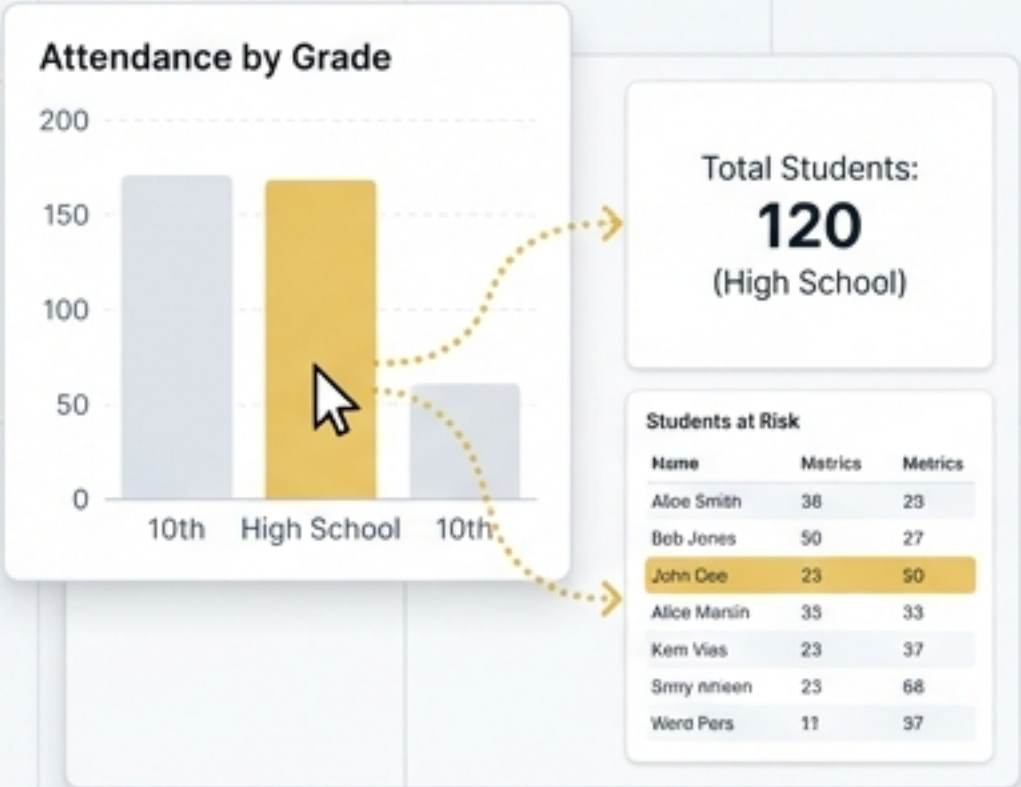
Cadence: Ensure all charts reflect the same time granularity (e.g., don't mix Daily vs Yearly metrics).

Interactivity: Cross-Filtering and Drill-Downs

1. Global Filter

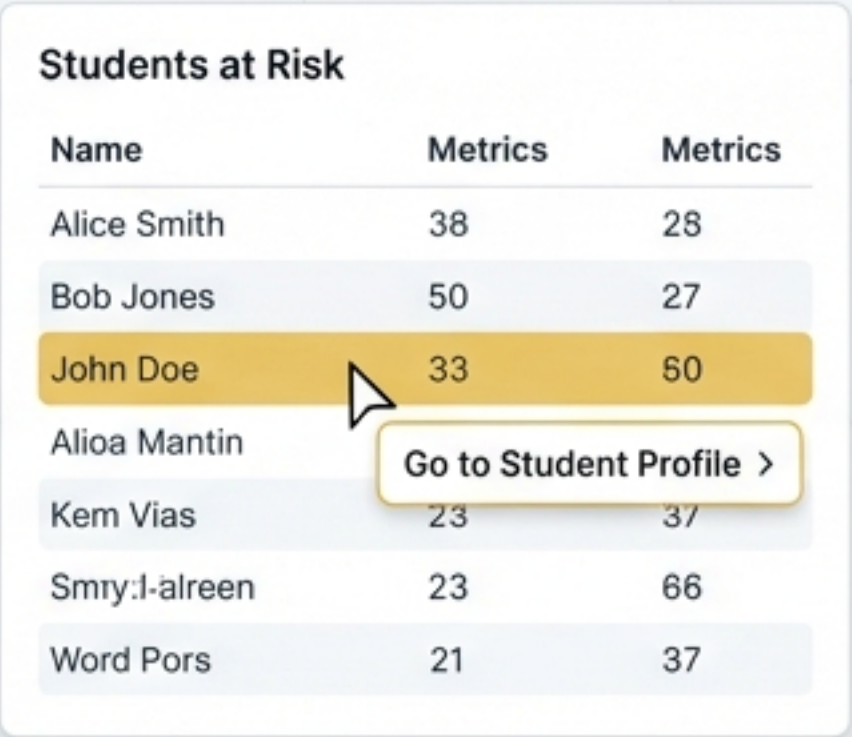


2. Cross-Filtering



Clicking "High School" filters all other charts to show only High School data.

3. Click Destination



Passes "student_id" parameter to external URL or detailed dashboard.

https://dashboard.com/student-profile?student_id=12345

Narrative Context: Using Markdown Text Cards

Weaving explanation directly into the dashboard.

Performance Report for {{School_Name}}

 ****Data Warning****: Attendance data is current as of yesterday.

****Methodology****: "Risk Score" is calculated using the 2026 OSS Framework algorithm.

Questions? Contact the Data Steward at data@district.edu

Dynamic Text: Use variables like {{School_Name}} so the narrative updates automatically based on dashboard filters.

Operations: Security and Subscriptions



Row-Level Security (RLS)



Teacher



Student_ID	Student_Name	Teacher_ID	Status
12345	Jane Doe	Me	Active
12345	Jane Doe	Me	Active
12346	#####	Other	Active
12347	#####	Other	Active
12348	#####	Other	Active
12349	#####	Other	Active
12340	#####	Other	Active



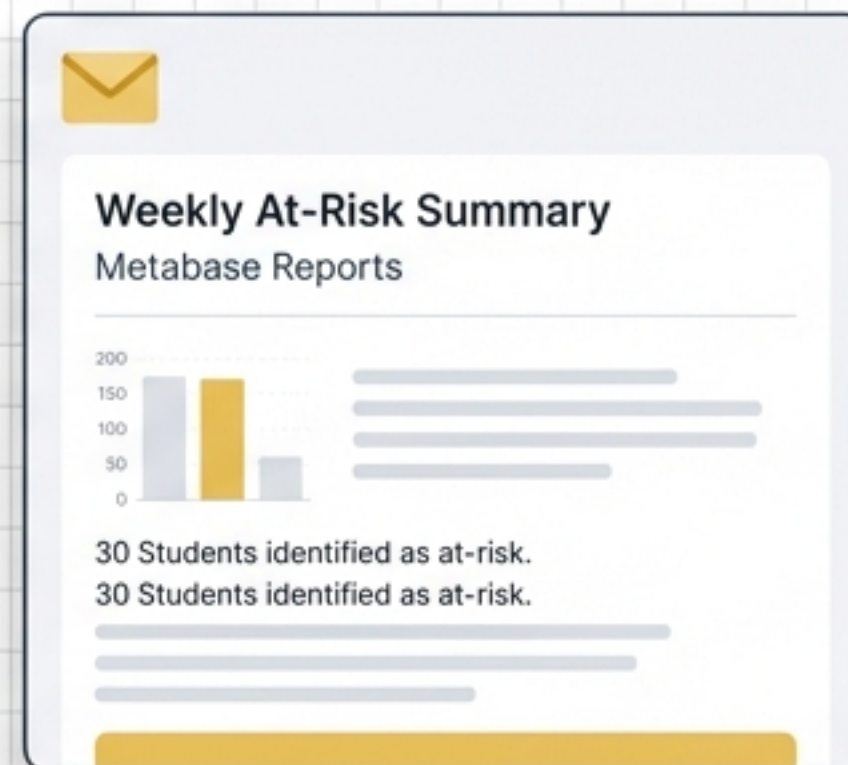
FERPA Compliance: Teachers see only their students.



Subscriptions (The Push Model)



08:00 AM



Automated Delivery: Don't wait for login.
Push data to users.